

Worked Answer Key

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1. $8 + 3 \times 4 = 8 + 12 = 20$.
2. $24 - 12 \div 3 = 24 - 4 = 20$.
3. $7 \times 5 + 6 = 35 + 6 = 41$.

Page 2

1. $(8 + 3) \times 4 = 11 \times 4 = 44$.
2. $24 - (12 \div 3) = 24 - 4 = 20$.
3. $7 \times (5 + 6) = 7 \times 11 = 77$.

Page 3

1. $36 \div 4 \times 3 = 9 \times 3 = 27$.
2. $5 \times 6 \div 2 = 30 \div 2 = 15$.
3. $48 \div 6 \times 2 + 1 = 8 \times 2 + 1 = 17$.

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1. $32 - 7 + 5 = 25 + 5 = 30$.
2. $18 + 9 - 6 = 27 - 6 = 21$.
3. $50 - 14 + 8 - 6 = 36 + 8 - 6 = 38$.

Page 5

1. $9 + 18 \div 3 = 9 + 6 = 15$.
2. $6 \times 7 - 20 = 42 - 20 = 22$.
3. $45 \div 5 + 4 \times 3 = 9 + 4 \times 3 = 9 + 12 = 21$.

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1. $3 \times (12 - 5) + 8 = 3 \times 7 + 8 = 21 + 8 = 29$.
2. $64 \div (4 + 4) + 7 = 64 \div 8 + 7 = 8 + 7 = 15$.
3. $40 - 2 \times (6 + 3) = 40 - 2 \times 9 = 40 - 18 = 22$.

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1. $4 \times 8 + 6 = 32 + 6 = 38$.
2. $72 \div 8 - 9 = 9 - 9 = 0$.
3. $(5 + 7) \times 3 = 12 \times 3 = 36$.

Page 8

1. $(6 + 4) \times 5 = 50$.
2. $(18 - 6) \div 3 = 4$.
3. $(2 + 8) \times 3 - 1 = 30 - 1 = 29$.

Page 9

1. $x = 65 - 27 = 38$; $38 + 27 = 65$.
2. $n = 300 - 148 = 152$; $148 + 152 = 300$.
3. $m = 1,000 - 425 = 575$; $575 + 425 = 1,000$.

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1. $y = 46 + 19 = 65$; $65 - 19 = 46$.
2. $p = 92 - 37 = 55$; $92 - 55 = 37$.
3. $q = 425 + 275 = 700$; $700 - 275 = 425$.

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1. $a = 36 \div 4 = 9$.
2. $b = 84 \div 7 = 12$.
3. $c = 144 \div 12 = 12$.

Page 12

1. $d = 9 \times 5 = 45$.
2. $e = 11 \times 8 = 88$.
3. $f = 156 \div 12 = 13$. Check: $12 \times 13 = 156$.

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1. $2x + 5 = 19$; $2x = 14$; $x = 7$. Check: $2 \times 7 + 5 = 14 + 5 = 19$.
2. $3y - 4 = 20$; $3y = 24$; $y = 8$. Check: $3 \times 8 - 4 = 24 - 4 = 20$.
3. $n \div 4 + 6 = 15$; $n \div 4 = 9$; $n = 36$. Check: $36 \div 4 + 6 = 9 + 6 = 15$.

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1. $(x + 3) \times 4 = 28$; $x + 3 = 7$; $x = 4$.
2. $5(p - 2) = 35$; $p - 2 = 7$; $p = 9$.
3. $q + 6 \times 7 = 80$; $q + 42 = 80$; $q = 38$.

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1. $x + 38 = 91$; $x = 53$.
2. $6p = 54$; $p = 9$ pencils per pack.
3. $x - 24 = 57$; $x = 81$.

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1. $4 \times 36 + 18 = 144 + 18 = 162$ students.
2. $(60 - 14) \div 2 = 46 \div 2 = 23$ dollars per jar.
3. $5 \times 24 - 17 = 120 - 17 = 103$ rolls.

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1. $6 + 4 \times 5 = 26$ and $(6 + 4) \times 5 = 50$; $26 < 50$.
2. $48 \div 6 + 2 = 10$ and $48 \div (6 + 2) = 6$; $10 > 6$.
3. $3 \times (9 - 2) = 21$ and $3 \times 9 - 2 = 25$; $21 < 25$.

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1. $7(20 + 3) = 7 \times 20 + 7 \times 3 = 140 + 21 = 161$.
2. Yes. Both equal 36 by the commutative property of multiplication.
3. $(12 + 3) \times 4 = 15 \times 4 = 60$. The parentheses make addition come first.

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1. $90 - 4 \times (8 + 2) = 90 - 4 \times 10 = 50$.
2. $3r + 7 = 31$; $3r = 24$; $r = 8$.
3. $7 \times 12 - 15 = 84 - 15 = 69$ members.

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1. Divide both sides by 5, then subtract 2: $5(x + 2) = 45$; $x + 2 = 9$; $x = 7$.
2. Answers vary. For example, $4 \times 9 - 4 = 36 - 4 = 32$ (multiply first) and $(6 + 2) \times 4 = 8 \times 4 = 32$ (add inside parentheses first).
3. Division comes before addition: $12 \div 4 = 3$, so $8 + 3 = 11$. Leo incorrectly added first.